सत्यमेव जयते
GOVERNMENT OF INDIA
MINISTRY OF SKILL DEVELOPMENT
& ENTREPRENEURSHIP

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कौशल गुणवत्ता प्रगति

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Qualification Pack

Geo-Spatial Surveying Supervisor

QP Code: SOI/AAS/Q0101

Version: 1.0

NSQF Level: 5

NATIONAL INSTITUTE FOR GEO-INFORMATICS SCIENCE AND TECHNOLOGY (NIGST) || NATIONAL INSTITUTE FOR GEO-INFORMATICS SCIENCE AND TECHNOLOGY , SURVEY OF INDIA , UPPAL HYDERABAD || email:nigst.hyd@surveyofindia.gov.in

Contents

SOI/AAS/Q0101: Geo-Spatial Surveying Supervisor

Brief Job Description

The individual will be capable of supervising data collection using advanced equipment such as Electronic Total Stations, GNSS, Digital Levels, and UAVs/Drones, and will ensure accurate data processing, visualization, and analysis using GIS software

Personal Attributes

The candidate should maintain a learning mindset and be prepared to supervise and manage team activities as part of their role.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. SOI/AAS/N0101: Supervise GNSS-based horizontal control surveys ensuring accuracy across all methods.
2. SOI/AAS/N0102: Oversee Total Station surveys for horizontal control and detail mapping.
3. SOI/AAS/N0103: Monitor vertical control surveys conducted using Digital Levels.
4. SOI/AAS/N0104: Coordinate and validate field data collection activities.
5. SOI/AAS/N0105: Supervise cadastral surveys ensuring spatial data accuracy.
6. SOI/AAS/N0106: Manage 3D data generation using UAV, photogrammetry, and LiDAR basics.
7. SOI/AAS/N0107: Ensure GIS maps are produced as per design parameters and templates.
8. DGT/VSQ/N0102: Employability Skills (60 Hours)

Qualification Pack (QP) Parameters

Sector	Aerospace and Aviation
Sub-Sector	Geo-Spatial
Occupation	Geo-Spatial Surveying Supervisor
Country	India

NSQF Level	5
Credits	20
Aligned to NCO/ISCO/ISIC Code	NCO 2165.0200 (Surveyor, Topographical) NCO 2165.0400 (Surveyor, Photogrammetry) NCO 2165.9900 (Cartographers and Surveyors, other)
Minimum Educational Qualification & Experience	12th grade pass with 2 year NTC/ CITS/NAC with 1 Year of experience OR 12th grade Pass with 4 Years of experience OR Completed 3 year diploma after 10th with 2 Years of experience OR Previous relevant Qualification of NSQF Level (Level 4.5) with 2 Years of experience OR Previous relevant Qualification of NSQF Level (Level 4) with 4 Years of experience
Minimum Level of Education for Training in School	12th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	18/07/2028
NSQC Approval Date	18/07/2025
Version	1.0
Reference code on NQR	QG-05-AA-04370-2025-V1-NIGST
NQR Version	1

SOI/AAS/N0101: Supervise GNSS-based horizontal control surveys ensuring accuracy across all methods.

Description

This unit describes the skills and knowledge required to provide horizontal control using GNSS technology through Static, Kinematic and other positioning methods, ensuring the specified accuracy and precision required for project work.

Scope

The scope covers the following :

- Control work by GNSS using various methods of survey i.e. Static and Kinematic.
- Control work Planning, data collection, data processing, adjustment & final report for horizontal control points.
- Used in surveying and mapping industry, construction industry.

Elements and Performance Criteria

Plan for GPS data collection.

To be competent, the user/individual on the job must be able to:

PC1. Choose the type of GNSS survey (Static or Kinematic) required based on project requirement.

PC2. Plan for observation of GNSS data and parameters (i.e. Network mode, Radial method, observation duration, observation time, elevation mask, etc.) to achieve the required accuracy.

PC3. Plan no. of resources required for the task i.e. instrument, software, manpower, vehicle etc.

PC4. Plan QA/QC planning for final horizontal control points.

Perform GNSS data collection.

To be competent, the user/individual on the job must be able to:

PC5. Collect master station data (previously established GPS points, Survey of India GPS library, CORS station data or other) to be used as control points as per plan.

PC6. Locate master station on ground and create marking (permanent or temporary as per requirement) for unknown points.

PC7. Setup GNSS instrument and measure height of the instrument and observe GNSS data as per plan.

PC8. Download observed data and save in the required format (native or RINEX) as per plan.

Process observed GNSS data.

To be competent, the user/individual on the job must be able to:

PC9. Setup software and input the observed GNSS data for processing

PC10. Carryout baseline measurement

PC11. Adjust baseline for co-ordinates

PC12. Generate final report of horizontal control of all points in the project.

PC13. Check accuracy requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Satellite Geodesy Keplers laws of satellite motion, Keplerian orbit in space, Perturbed satellite orbit, Gravitational perturbation, Atmospheric drag, Solar radiation pressure etc.

KU2. Basics of GNSS, Various GNSS systems such as GPS, GLONASS, GALELIO, Beidou Fundamentals of GNSS, Definitions of the terms used in GNSS technology, GNSS signal characteristics. GNSS observables Code and Pseudo Range measurements, Broad cast Navigation message, Ephemeris, Carrier waves and Carrier Phase measurement. Data formats like RINEX etc., RTK Link Options, RTCM data Transmission, Types of GNSS receivers and antennas.

KU3. Differencing of GNSS data and parameters estimation, Single differences, Double differences and Triple differences, Solution of Ambiguities, Possible application of GNSS.

KU4. Plan and execute the horizontal control survey by Various positioning techniques in GNSS such as Static, Rapid-static, Differential GNSS, kinematic and stop and go surveying technique, Network RTK

KU5. GNSS signal propagation and error budget, Satellite Geometry and Accuracy measures. Ionospheric and Tropospheric delay, Basics of theory of error for better understanding about accuracy, error, adjustment and chi square test for QA/QC check of data processing report.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Plan horizontal control with GNSS for required accuracy.

GS2. Execution of the GNSS data observation and processing of the data.

GS3. Optimum utilization of time and resources within the estimated budget.

GS4. Troubleshooting in case the required accuracy could not be achieved in the first attempt and take corrective decision and action.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Plan for GPS data collection.	**30**	**-**	**-**	**-**
PC1. Choose the type of GNSS survey (Static or Kinematic) required based on project requirement.	10	-	-	-
PC2. Plan for observation of GNSS data and parameters (i.e. Network mode, Radial method, observation duration, observation time, elevation mask, etc.) to achieve the required accuracy.	10	-	-	-
PC3. Plan no. of resources required for the task i.e. instrument, software, manpower, vehicle etc.	5	-	-	-
PC4. Plan QA/QC planning for final horizontal control points.	5	-	-	-
Perform GNSS data collection.	**4**	**15**	**-**	**-**
PC5. Collect master station data (previously established GPS points, Survey of India GPS library, CORS station data or other) to be used as control points as per plan.	2	2	-	-
PC6. Locate master station on ground and create marking (permanent or temporary as per requirement) for unknown points.	2	2	-	-
PC7. Setup GNSS instrument and measure height of the instrument and observe GNSS data as per plan.	-	6	-	-
PC8. Download observed data and save in the required format (native or RINEX) as per plan.	-	5	-	-
Process observed GNSS data.	**16**	**35**	**-**	**-**
PC9. Setup software and input the observed GNSS data for processing	2	5	-	-
PC10. Carryout baseline measurement	5	10	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Adjust baseline for co-ordinates	5	10	-	-
PC12. Generate final report of horizontal control of all points in the project.	2	5	-	-
PC13. Check accuracy requirement	2	5	-	-
NOS Total	**50**	**50**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	SOI/AAS/N0101
NOS Name	Supervise GNSS-based horizontal control surveys ensuring accuracy across all methods.
Sector	Aerospace and Aviation
Sub-Sector	
Occupation	Geo-Spatial Surveying Supervisor
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	18/07/2025
Next Review Date	18/07/2028
NSQC Clearance Date	18/07/2025

SOI/AAS/N0102: Oversee Total Station surveys for horizontal control and detail mapping.

Description

This unit describes to provide horizontal control using Electronic Total Station through traversing method for a specified accuracy/precision required for a project work. It also describes to provide detail survey using Electronic Total Station for mapping of details related to a project requirement inside project area to create a map/GIS data.

Scope

The scope covers the following :

- Traverse planning, execution, data adjustment and final report for control work using Total Station within the specified limit.
- Detail survey using Total Station for mapping purpose. Various feature details are mapped and plotted to create a map.
- Used in surveying and mapping industry, construction industry, asset management.

Elements and Performance Criteria

Plan for Traverse using Total Station.

To be competent, the user/individual on the job must be able to:

PC1. Plan traverse to provide or extend the control point at a specified location.

PC2. Plan number of resources required for the task i.e. instrument, software, manpower, vehicle, etc.

PC3. Do QA/QC planning for final horizontal control points.

Data collection using Total Station.

To be competent, the user/individual on the job must be able to:

PC4. Identify two control stations whose co-ordinates are known for setup of Total Station or create two using GNSS survey method.

PC5. Setup Total Station using known control points.

PC6. Start traversing and create temporary points in between and close the traverse at the same point or some known point.

Traverse adjustment.

To be competent, the user/individual on the job must be able to:

PC7. Perform traverse adjustment and generate report for final control value of unknown point within the permissible closing error limit.

PC8. Submit a report with new control point coordinates, proper sketch of traverse, error report etc

Plan for Detail Survey using Total Station.

To be competent, the user/individual on the job must be able to:

PC9. Plan for detail survey i.e. list out the details which need to be picked. Plan for nearby provisioning of control points.

PC10. Plan number of resources required for the task i.e. instrument, software, manpower, vehicle, etc.

PC11. Plan QA/QC procedure.

Data collection using Total Station.

To be competent, the user/individual on the job must be able to:

PC12. Identify two intervisible control stations whose co-ordinates are known for setup of Total Station or create two using GNSS survey method.

PC13. Setup Total Station using known control points.

PC14. Start survey to pick up the details of the features, collect and store the data as per plan.

Download and data representation.

To be competent, the user/individual on the job must be able to:

PC15. Download data and save.

PC16. Prepare a map with necessary map template and legends.

Design and calculate the various elements of curves.

To be competent, the user/individual on the job must be able to:

PC17. Calculate the various elements of horizontal and vertical curves.

PC18. Calculate the required elements for setting out of curve.

PC19. Select the proper method to stake out the curve / tunnel alignment as per design.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Working principle of Electronic Total Station, Basic measurement involved i.e. angular and linear measurement

KU2. Basics of control survey using Electronic Total Station. Its planning, execution and final report generation

KU3. Basics of detail survey using Electronic Total Station. Its planning, execution and final report generation

KU4. The basics of curve design and its elements for stake out of curve at field

KU5. Types of error, their adjustment and precaution during data collection involved in traversing using total station.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Plan horizontal control using total station of required accuracy.

GS2. Execution of the control work using total station, data processing and final report generation of the data.

GS3. Plan detail survey of required accuracy.

GS4. Execution of detail survey using total station, data processing and final map generation of the data.

GS5. Application of surveying and mapping principle for setting out of curve and tunnel alignment.

GS6. Optimum utilization of time and resources within the estimated budget.

GS7. Troubleshooting in case the required accuracy could not be achieved in the first attempt and take corrective action and decision.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Plan for Traverse using Total Station.	**15**	**-**	**-**	**-**
PC1. Plan traverse to provide or extend the control point at a specified location.	5	-	-	-
PC2. Plan number of resources required for the task i.e. instrument, software, manpower, vehicle, etc.	5	-	-	-
PC3. Do QA/QC planning for final horizontal control points.	5	-	-	-
Data collection using Total Station.	**6**	**14**	**-**	**-**
PC4. Identify two control stations whose co-ordinates are known for setup of Total Station or create two using GNSS survey method.	2	2	-	-
PC5. Setup Total Station using known control points.	2	2	-	-
PC6. Start traversing and create temporary points in between and close the traverse at the same point or some known point.	2	10	-	-
Traverse adjustment.	**8**	**10**	**-**	**-**
PC7. Perform traverse adjustment and generate report for final control value of unknown point within the permissible closing error limit.	5	5	-	-
PC8. Submit a report with new control point coordinates, proper sketch of traverse, error report etc	3	5	-	-
Plan for Detail Survey using Total Station.	**9**	**-**	**-**	**-**
PC9. Plan for detail survey i.e. list out the details which need to be picked. Plan for nearby provisioning of control points.	5	-	-	-
PC10. Plan number of resources required for the task i.e. instrument, software, manpower, vehicle, etc.	2	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Plan QA/QC procedure.	2	-	-	-
Data collection using Total Station.	**3**	**14**	**-**	**-**
PC12. Identify two intervisible control stations whose co-ordinates are known for setup of Total Station or create two using GNSS survey method.	2	2	-	-
PC13. Setup Total Station using known control points.	1	2	-	-
PC14. Start survey to pick up the details of the features, collect and store the data as per plan.	-	10	-	-
Download and data representation.	**4**	**12**	**-**	**-**
PC15. Download data and save.	1	2	-	-
PC16. Prepare a map with necessary map template and legends.	3	10	-	-
Design and calculate the various elements of curves.	**5**	**-**	**-**	**-**
PC17. Calculate the various elements of horizontal and vertical curves.	2	-	-	-
PC18. Calculate the required elements for setting out of curve.	2	-	-	-
PC19. Select the proper method to stake out the curve / tunnel alignment as per design.	1	-	-	-
NOS Total	**50**	**50**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	SOI/AAS/N0102
NOS Name	Oversee Total Station surveys for horizontal control and detail mapping.
Sector	Aerospace and Aviation
Sub-Sector	
Occupation	Geo-Spatial Surveying Supervisor
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	18/07/2025
Next Review Date	18/07/2028
NSQC Clearance Date	18/07/2025

SOI/AAS/N0103: Monitor vertical control surveys conducted using Digital Levels.

Description

This unit describes to provide vertical control using Digital Level for a specified accuracy/precision required for a project work.

Scope

The scope covers the following :

- Level network planning, data collection, data adjustment and final production of vertical control value i.e. height.
- Used for surveying and mapping industry, water resource management agencies, other related agencies where elevation data is required for hydrological surveys.

Elements and Performance Criteria

Plan for vertical control using digital level.

To be competent, the user/individual on the job must be able to:

PC1. Plan vertical control survey route to connect the unknown height point from known point.

PC2. Plan for gravity data observation.

PC3. Plan no. of resources required for the task i.e. instrument, software, manpower, vehicle etc.

PC4. Plan QA/QC planning for final horizontal control points.

Data collection using Digital Level.

To be competent, the user/individual on the job must be able to:

PC5. Check digital level calibration and collimation error and check the suitability of the instrument to start the work.

PC6. Start levelling from known/unknown points along the route, record data digitally or physically, create temporary bench mark and reach to the final bench mark as per planning.

PC7. Enter permanent and temporary benchmark details (i.e. written description, sketches, co-ordinates) used/created during levelling in a register as per provide format.

Data collection using gravimeter.

To be competent, the user/individual on the job must be able to:

PC8. Check calibration and other parameters for suitability of instrument to observe gravity observation.

PC9. Collect, record and store gravity data on the levelling bench marks.

Data download

To be competent, the user/individual on the job must be able to:

PC10. Download levelling data and save in required format.

PC11. Download gravity data and save it in required format.

Data checking and adjustment.

To be competent, the user/individual on the job must be able to:

PC12. Check levelling data to meet the specified criteria and provide a decision for acceptance or re-levelling

PC13. Check gravity data as per standard procedure and perform necessary correction.

PC14. Perform adjustment operation using accepted levelling and gravity data to obtain the final value of control points i.e. height.

PC15. Create final report for vertical control survey with final height of unknown stations with details of raw data collected, corrections and adjustments performed and details of temporary station with sketches as per standard practice.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. The Earth and its gravity field, gravity, Curvature of Level surfaces and Plumb line

KU2. Principle of levelling, classification accuracy standard, various methods for providing vertical control i.e. High precision levelling, ST/DT levelling.

KU3. Vertical Datum: Mean Sea Level, Brief history of Indian Levelling network, Different height systems, Orthometric height, Ellipsoidal height, Geo-potential Number, Redefinition of Indian vertical datum

KU4. Working principle of Digital level instrument, sources of error, testing, adjustment and handling.

KU5. Working principle of Gravimeter, sources of error, testing, adjustment and handling.

KU6. Final computation and Adjustment of levelling and Gravity network.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Vertical control using digital level of required accuracy.

GS2. Execution of the control work for height using digital level, data processing and final report generation of the data.

GS3. Optimum utilization of time and resources within the estimated budged.

GS4. Troubleshooting in case the required accuracy could not be achieved in the first attempt and take corrective action and decision.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Plan for vertical control using digital level.	**14**	**-**	**-**	**-**
PC1. Plan vertical control survey route to connect the unknown height point from known point.	4	-	-	-
PC2. Plan for gravity data observation.	4	-	-	-
PC3. Plan no. of resources required for the task i.e. instrument, software, manpower, vehicle etc.	3	-	-	-
PC4. Plan QA/QC planning for final horizontal control points.	3	-	-	-
Data collection using Digital Level.	**6**	**20**	**-**	**-**
PC5. Check digital level calibration and collimation error and check the suitability of the instrument to start the work.	2	5	-	-
PC6. Start levelling from known/unknown points along the route, record data digitally or physically, create temporary bench mark and reach to the final bench mark as per planning.	2	10	-	-
PC7. Enter permanent and temporary benchmark details (i.e. written description, sketches, co-ordinates) used/created during levelling in a register as per provide format.	2	5	-	-
Data collection using gravimeter.	**4**	**10**	**-**	**-**
PC8. Check calibration and other parameters for suitability of instrument to observe gravity observation.	2	5	-	-
PC9. Collect, record and store gravity data on the levelling bench marks.	2	5	-	-
Data download	**4**	**4**	**-**	**-**
PC10. Download levelling data and save in required format.	2	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Download gravity data and save it in required format.	2	2	-	-
Data checking and adjustment.	**22**	**16**	**-**	**-**
PC12. Check levelling data to meet the specified criteria and provide a decision for acceptance or re-levelling	5	5	-	-
PC13. Check gravity data as per standard procedure and perform necessary correction.	5	3	-	-
PC14. Perform adjustment operation using accepted levelling and gravity data to obtain the final value of control points i.e. height.	10	5	-	-
PC15. Create final report for vertical control survey with final height of unknown stations with details of raw data collected, corrections and adjustments performed and details of temporary station with sketches as per standard practice.	2	3	-	-
NOS Total	**50**	**50**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	SOI/AAS/N0103
NOS Name	Monitor vertical control surveys conducted using Digital Levels.
Sector	Aerospace and Aviation
Sub-Sector	
Occupation	Geo-Spatial Surveying Supervisor
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	18/07/2025
Next Review Date	18/07/2028
NSQC Clearance Date	18/07/2025

SOI/AAS/N0104: Coordinate and validate field data collection activities.

Description

This unit describes to carry out field data collection for features and attribute using FDC.

Scope

The scope covers the following :

- Field data collection for features and attributes for GIS ready maps. Planning, execution and quality control of the data collected.
- Used for surveying and mapping industry, construction industry, asset management of any entity.

Elements and Performance Criteria

Plan for feature and attribute collection.

To be competent, the user/individual on the job must be able to:

PC1. Plan for data collection (i.e. design of forms, fields and other necessary data required), store, and representation as per the user requirement.

PC2. Plan for base map, WFS or vector data for feature or attribute data collection.

PC3. Plan number of resources required for the task i.e. instrument, software, manpower, vehicle etc

PC4. Plan QA/QC procedure.

Data collection using FDC.

To be competent, the user/individual on the job must be able to:

PC5. Prepare base map, WFS or vector data for feature or attribute data collection.

PC6. Setup FDC with all basic data stored or through web services.

PC7. Connect external GNSS for better accuracy if provisioned during planning

PC8. Start survey to pick up the details of the features and fill their attribute in the designed fields. In case feature is already there fill up the attribute details in the respective fields.

Download and data representation.

To be competent, the user/individual on the job must be able to:

PC9. Check the feature and attribute data for its consistency with earlier data available and proceed as per the policy of data acceptance.

PC10. Download data and save.

PC11. Prepare a map with necessary map template and legends as per plan.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Workflow for Collection of data using Field Data Collection (FDC) devices.

KU2. Basics of base map service (i.e. WFS, WMS, WCS, WMTS) and its uses and limitations

KU3. Basic understanding of feature and attribute fields as per project requirement

KU4. Data acceptance policy in case where previous data is also available and reporting of the same.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Plan and execution of feature and attribute collection for client requirement.

GS2. Optimum utilization of time and resources within the estimated budged.

GS3. Troubleshooting in case the required accuracy could not be achieved in the first attempt and take corrective action and decision.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Plan for feature and attribute collection.	**25**	**-**	**-**	**-**
PC1. Plan for data collection (i.e. design of forms, fields and other necessary data required), store, and representation as per the user requirement.	10	-	-	-
PC2. Plan for base map, WFS or vector data for feature or attribute data collection.	5	-	-	-
PC3. Plan number of resources required for the task i.e. instrument, software, manpower, vehicle etc	5	-	-	-
PC4. Plan QA/QC procedure.	5	-	-	-
Data collection using FDC.	**15**	**30**	**-**	**-**
PC5. Prepare base map, WFS or vector data for feature or attribute data collection.	5	10	-	-
PC6. Setup FDC with all basic data stored or through web services.	5	5	-	-
PC7. Connect external GNSS for better accuracy if provisioned during planning	2	5	-	-
PC8. Start survey to pick up the details of the features and fill their attribute in the designed fields. In case feature is already there fill up the attribute details in the respective fields.	3	10	-	-
Download and data representation.	**10**	**20**	**-**	**-**
PC9. Check the feature and attribute data for its consistency with earlier data available and proceed as per the policy of data acceptance.	5	5	-	-
PC10. Download data and save.	2	5	-	-
PC11. Prepare a map with necessary map template and legends as per plan.	3	10	-	-
NOS Total	**50**	**50**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	SOI/AAS/N0104
NOS Name	Coordinate and validate field data collection activities.
Sector	Aerospace and Aviation
Sub-Sector	
Occupation	Geo-Spatial Surveying Supervisor
NSQF Level	5
Credits	1
Version	1.0
Last Reviewed Date	18/07/2025
Next Review Date	18/07/2028
NSQC Clearance Date	18/07/2025

SOI/AAS/N0105: Supervise cadastral surveys ensuring spatial data accuracy.

Description

This unit describes to carry out survey activity related to cadastral mapping in India. Create, update and relay of survey details. Read the legacy/ current cadastral maps and records to work with it.

Scope

The scope covers the following :

- Basics about cadastral maps, maintenance of cadastral records, conversion of cadastral record in GIS format.
- Use of latest technology (photogrammetry, Remote sensing, UAV) in cadastral survey and Land information system.
- Various government initiatives for modernization of land records in India.
- Used for surveying and mapping industry, Central and state government agencies related land record departments.

Elements and Performance Criteria

Read, Create & modify cadastral maps and cadastral record.

To be competent, the user/individual on the job must be able to:

PC1. Read the cadastral maps and other record.

PC2. Measure distance using cadastral maps.

PC3. Relaying the missing pillar based on survey data.

Transform/convert cadastral map into GIS ready map.

To be competent, the user/individual on the job must be able to:

PC4. Scan the existing cadastral map.

PC5. Georeference the scanned cadastral map using ortho rectified image (ORI) from drone/UAV, Satellite etc. or ground control points.

PC6. Digitize the georeferenced map and prepare a GIS ready map with attribute field as per project requirement.

PC7. Entry of attribute data into existing field.

Use of UAV for Cadastral survey.

To be competent, the user/individual on the job must be able to:

PC8. Plan for Cadastral survey using UAV to generate ORI.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Basics of Cadastral survey, mapping and its record.

KU2. Read and interpret the details from cadastral record for relaying, updating, etc.
KU3. Use of latest technology to create, update/modify the cadastral records.
KU4. Common Terminology Used in Cadastral Survey in India

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Read and understand the cadastral record.
GS2. Basic workflow adopted in creation of cadastral record. Various basic administrative and legal aspects related to cadastral map.
GS3. Optimum utilization of time and resources within the estimated budged.
GS4. Troubleshooting in case the required accuracy could not be achieved in the first attempt and take corrective action and decision.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Read, Create & modify cadastral maps and cadastral record.	**20**	**20**	**-**	**-**
PC1. Read the cadastral maps and other record.	10	5	-	-
PC2. Measure distance using cadastral maps.	5	5	-	-
PC3. Relaying the missing pillar based on survey data.	5	10	-	-
Transform/convert cadastral map into GIS ready map.	**25**	**30**	**-**	**-**
PC4. Scan the existing cadastral map.	5	5	-	-
PC5. Georeference the scanned cadastral map using ortho rectified image (ORI) from drone/UAV, Satellite etc. or ground control points.	10	10	-	-
PC6. Digitize the georeferenced map and prepare a GIS ready map with attribute field as per project requirement.	5	10	-	-
PC7. Entry of attribute data into existing field.	5	5	-	-
Use of UAV for Cadastral survey.	**5**	**-**	**-**	**-**
PC8. Plan for Cadastral survey using UAV to generate ORI.	5	-	-	-
NOS Total	**50**	**50**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	SOI/AAS/N0105
NOS Name	Supervise cadastral surveys ensuring spatial data accuracy.
Sector	Aerospace and Aviation
Sub-Sector	
Occupation	Geo-Spatial Surveying Supervisor
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	18/07/2025
Next Review Date	18/07/2028
NSQC Clearance Date	18/07/2025

SOI/AAS/N0106: Manage 3D data generation using UAV, photogrammetry, and LiDAR basics.

Description

This unit describes to perform basic operations to generate Digital Surface model (DSM), Digital Elevation Model (DEM) & Orthorectified Image (ORI) using photogrammetric method and UAV (Drones). Unit also include the feature extraction from 3D datasets. Regulatory aspects related to drone data capture is also introduced in the unit.

Scope

The scope covers the following :

- Basic planning and understanding of photogrammetry.
- Mission planning for drone data capture, execution of the flight.
- Processing of captured data to generate the final products like DSM, DEM, ORI and 3D models.
- Used for surveying and mapping in several sectors.
- Basic of LiDAR remote sensing and its component.

Elements and Performance Criteria

Plan for capture of data (flight planning).

To be competent, the user/individual on the job must be able to:

PC1. Plan for data capture element like front overlap, side overlap, flight height, GSD based on project requirement.

PC2. Plan for no. of ground control points and check points required for the georeferencing and QA/QC

PC3. Plan QA/QC planning for final horizontal control points.

PC4. Plan no. of resources required for the task i.e. instrument, software, manpower, vehicle etc.

Perform Aerial triangulation.

To be competent, the user/individual on the job must be able to:

PC5. Geo-indexing of photos: Creation of photo ties, examination of overlaps, flight direction control planning on index

PC6. Create Block file for the captured data.

PC7. Generate manual & automatic tie-points, measure GCP

PC8. Perform Bundle Block adjustment

Create DSM, DEM, Contour and ORI.

To be competent, the user/individual on the job must be able to:

PC9. Generate DSM as per project requirement.

PC10. Generate DEM as per project requirement.

PC11. Generate Contour as per project requirement.

PC12. Generate ORI as per project requirement.

Extract 3D feature.

To be competent, the user/individual on the job must be able to:

PC13. Set up of system for 3D feature extraction like setup of topo mouse, software etc

PC14. Extract 3D feature as per project requirement and save the data in required format.

Mission Planning for Drone/UAV.

To be competent, the user/individual on the job must be able to:

PC15. Plan mission to capture the data.

PC16. Plan for no. of ground control points and check points required for the georeferencing and QA/QC

PC17. Plan QA/QC method for horizontal control points.

PC18. Plan no. of resources required for the task i.e. instrument, software, manpower, vehicle etc.

Execute flight for data acquisition.

To be competent, the user/individual on the job must be able to:

PC19. Perform preflight checks including regulatory permission.

PC20. Execute flight as per flight plan

PC21. Download flight data along with other necessary files i.e. flight logs, base station data.

PC22. Fill up all required/mandatory formats.

Process acquired data.

To be competent, the user/individual on the job must be able to:

PC23. Geotag the drone data.

PC24. Generate DSM and export as per project requirement.

PC25. Generate DEM and export as per project requirement.

PC26. Generate ORI and export as per project requirement.

PC27. Generate Contour and export as per project requirement.

PC28. Generate 3D model and export as per project requirement.

Mission Planning (LiDAR)

To be competent, the user/individual on the job must be able to:

PC29. Plan mission to capture the data.

PC30. Plan for no. of ground control points and check points required for the georeferencing and QA/QC

PC31. Plan QA/QC method for horizontal control points.

PC32. Plan no. of resources required for the task i.e. instrument, software, manpower, vehicle etc.

Open Lidar data in software.

To be competent, the user/individual on the job must be able to:

PC33. Open LAS data in software

PC34. Measure height from LAS data.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Principles of photogrammetry, aerial photography and its components, data processing of aerial photographs.

KU2. Project planning for aerial data capturing. Ground control requirement for georeferencing and Quality control to meet the project requirement.

KU3. Data processing steps like Aerial triangulation, bundle block adjustment. Generation and quality control of DSM, DEM, Contour & ORI.

KU4. Setup requirement for 3D feature extraction and its components.

KU5. Principles of Drone photogrammetry. Types of Drones, its components and data processing of raw data captured.

KU6. Project planning for drone data capturing. Ground control requirement for georeferencing and Quality control to meet the project requirement.

KU7. Data processing steps like tie point generation, spares point cloud, dense point cloud Aerial triangulation, bundle block adjustment.

KU8. Generation and quality control of DSM, DEM, Contour, ORI & 3D model.

KU9. Basic Measurement Principles & Basic Physics of Laser Ranging, Basic Components of ALS Technology and functioning, Laser Beam and its interaction with targets, Discrete and Full Waveform Laser Scan Data, LASER Classification- Eye Safety, Class I to Class IV Lasers.

KU10. Flight Planning & Ground Controls, Guidelines, Best Practices, Standards for LiDAR Data Collection, LiDAR direct geo-referencing, using raw LIDAR file, trajectory and base station.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Plan for aerial data capture and processing of the data to generate the final deliverables.

GS2. Control provision for positional accuracy and quality control plan to meet the project requirement.

GS3. Optimum utilization of time and resources within the estimated budged.

GS4. Troubleshooting in case the required accuracy could not be achieved in the first attempt and take corrective action and decision.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Plan for capture of data (flight planning).	**8**	**-**	**-**	**-**
PC1. Plan for data capture element like front overlap, side overlap, flight height, GSD based on project requirement.	2	-	-	-
PC2. Plan for no. of ground control points and check points required for the georeferencing and QA/QC	2	-	-	-
PC3. Plan QA/QC planning for final horizontal control points.	2	-	-	-
PC4. Plan no. of resources required for the task i.e. instrument, software, manpower, vehicle etc.	2	-	-	-
Perform Aerial triangulation.	**8**	**7**	**-**	**-**
PC5. Geo-indexing of photos: Creation of photo ties, examination of overlaps, flight direction control planning on index	2	-	-	-
PC6. Create Block file for the captured data.	2	3	-	-
PC7. Generate manual & automatic tie-points, measure GCP	2	2	-	-
PC8. Perform Bundle Block adjustment	2	2	-	-
Create DSM, DEM, Contour and ORI.	**8**	**8**	**-**	**-**
PC9. Generate DSM as per project requirement.	2	2	-	-
PC10. Generate DEM as per project requirement.	2	2	-	-
PC11. Generate Contour as per project requirement.	2	2	-	-
PC12. Generate ORI as per project requirement.	2	2	-	-
Extract 3D feature.	**5**	**4**	**-**	**-**
PC13. Set up of system for 3D feature extraction like setup of topo mouse, software etc	3	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. Extract 3D feature as per project requirement and save the data in required format.	2	2	-	-
Mission Planning for Drone/UAV.	**7**	**3**	**-**	**-**
PC15. Plan mission to capture the data.	2	2	-	-
PC16. Plan for no. of ground control points and check points required for the georeferencing and QA/QC	2	1	-	-
PC17. Plan QA/QC method for horizontal control points.	2	-	-	-
PC18. Plan no. of resources required for the task i.e. instrument, software, manpower, vehicle etc.	1	-	-	-
Execute flight for data acquisition.	**2**	**12**	**-**	**-**
PC19. Perform preflight checks including regulatory permission.	1	3	-	-
PC20. Execute flight as per flight plan	1	5	-	-
PC21. Download flight data along with other necessary files i.e. flight logs, base station data.	-	2	-	-
PC22. Fill up all required/mandatory formats.	-	2	-	-
Process acquired data.	**6**	**12**	**-**	**-**
PC23. Geotag the drone data.	1	2	-	-
PC24. Generate DSM and export as per project requirement.	1	2	-	-
PC25. Generate DEM and export as per project requirement.	1	2	-	-
PC26. Generate ORI and export as per project requirement.	1	2	-	-
PC27. Generate Contour and export as per project requirement.	1	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC28. Generate 3D model and export as per project requirement.	1	2	-	-
Mission Planning (LiDAR)	**4**	**-**	**-**	**-**
PC29. Plan mission to capture the data.	1	-	-	-
PC30. Plan for no. of ground control points and check points required for the georeferencing and QA/QC	1	-	-	-
PC31. Plan QA/QC method for horizontal control points.	1	-	-	-
PC32. Plan no. of resources required for the task i.e. instrument, software, manpower, vehicle etc.	1	-	-	-
Open Lidar data in software.	**2**	**4**	**-**	**-**
PC33. Open LAS data in software	1	2	-	-
PC34. Measure height from LAS data.	1	2	-	-
NOS Total	**50**	**50**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	SOI/AAS/N0106
NOS Name	Manage 3D data generation using UAV, photogrammetry, and LiDAR basics.
Sector	Aerospace and Aviation
Sub-Sector	
Occupation	Geo-Spatial Surveying Supervisor
NSQF Level	5
Credits	4
Version	1.0
Last Reviewed Date	18/07/2025
Next Review Date	18/07/2028
NSQC Clearance Date	18/07/2025

SOI/AAS/N0107: Ensure GIS maps are produced as per design parameters and templates.

Description

This unit describes the basic principle and workflow required to create a GIS map from beginning as per project requirements.

Scope

The scope covers the following :

- Fundamental of Cartography for map production and its implementation through GIS software.
- Fundamental of Geographical Information System (GIS).
- Used for surveying and mapping industry, Land management authorities etc.

Elements and Performance Criteria

Download and use open-source software and data.

To be competent, the user/individual on the job must be able to:

PC1. Search download, install, open & utilize the open-source GIS software

PC2. Download open-source and freely available raster and vector data from various sources like Survey of India, Bhuvan, USGS, Google earth, Street map etc

Geo-reference the raster data.

To be competent, the user/individual on the job must be able to:

PC3. Open software and load the raster image/satellite image/ORI.

PC4. Identify and mark the control point on input dataset and enter the known co-ordinate value.

PC5. Geo-reference the input dataset using a proper georeferencing method.

PC6. Calculate the RMSE value of final geo-referencing process and check it against the permissible value.

Create GIS file and spatial database.

To be competent, the user/individual on the job must be able to:

PC7. Create a file geodatabse/geopackage/shapefile and save it.

PC8. Install database software for geospatial data, create tables, connect it with GIS software.

PC9. Create Schema including Domains & Sub-types to implement Spatial Data Model Structure (SDMS) as per project requirement.

PC10. Create new stylesheet and Customized symbols in a style-sheet using font symbols in third-party s/w, Configuring the Colour Table as per SOI/project required symbology

PC11. Install SOI/project required, font-files and Attach SOI/project required Style-sheet.

Extract feature from geo-referenced raster dataset as per SDMS with proper symbology.

To be competent, the user/individual on the job must be able to:

PC12. Interpret feature, choose right feature class/category.

PC13. Digitize and attach proper symbology.

QA/QC of digitized data before map finalization.

To be competent, the user/individual on the job must be able to:

PC14. Create rule-based topology.

PC15. Run topological validation.

PC16. Correct error after topological validation, if any

Prepare map layout as per template if any and print the map.

To be competent, the user/individual on the job must be able to:

PC17. Create map layout in software as per project map template.

PC18. Create pdf file for printing in large format printers.

QA/QC of printed map

To be competent, the user/individual on the job must be able to:

PC19. Examine the hard copy of map for omissions and mistakes. Check cartographic details and overlapping symbols, text placement and spelling correction etc.

Create thematic map

To be competent, the user/individual on the job must be able to:

PC20. Create a large-scale thematic map as per project requirement.

Transform Datum from one datum to another datum.

To be competent, the user/individual on the job must be able to:

PC21. Perform datum transformation using 7 parameters.

Convert co-ordinate from one system to another.

To be competent, the user/individual on the job must be able to:

PC22. Perform co-ordinate transformation from one system to another.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Fundamentals of Cartography for map design. Basics of GIS and various open-source software and data available

KU2. Various map element like Datum, projection, symbology, topology, map layout and printing.

KU3. GIS software tools and functioning to implement the theoretical concepts.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Plan for various resources like base image, control point for referencing, software to be used, map template requirement and QA/QC aspect of the data before finalization of the map.

GS2. Execute, supervise and manage the map creation related projects.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Download and use open-source software and data.	**4**	**4**	**-**	**-**
PC1. Search download, install, open & utilize the open-source GIS software	2	2	-	-
PC2. Download open-source and freely available raster and vector data from various sources like Survey of India, Bhuvan, USGS, Google earth, Street map etc	2	2	-	-
Geo-reference the raster data.	**12**	**9**	**-**	**-**
PC3. Open software and load the raster image/satellite image/ORI.	2	2	-	-
PC4. Identify and mark the control point on input dataset and enter the known co-ordinate value.	-	2	-	-
PC5. Geo-reference the input dataset using a proper georeferencing method.	5	3	-	-
PC6. Calculate the RMSE value of final geo-referencing process and check it against the permissible value.	5	2	-	-
Create GIS file and spatial database.	**16**	**13**	**-**	**-**
PC7. Create a file geodatabse/geopackage/shapefile and save it.	5	2	-	-
PC8. Install database software for geospatial data, create tables, connect it with GIS software.	4	3	-	-
PC9. Create Schema including Domains & Sub-types to implement Spatial Data Model Structure (SDMS) as per project requirement.	5	3	-	-
PC10. Create new stylesheet and Customized symbols in a style-sheet using font symbols in third-party s/w, Configuring the Colour Table as per SOI/project required symbology	2	3	-	-
PC11. Install SOI/project required, font-files and Attach SOI/project required Style-sheet.	-	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Extract feature from geo-referenced raster dataset as per SDMS with proper symbology.	**5**	**5**	**-**	**-**
PC12. Interpret feature, choose right feature class/category.	3	2	-	-
PC13. Digitize and attach proper symbology.	2	3	-	-
QA/QC of digitized data before map finalization.	**3**	**6**	**-**	**-**
PC14. Create rule-based topology.	3	3	-	-
PC15. Run topological validation.	-	1	-	-
PC16. Correct error after topological validation, if any	-	2	-	-
Prepare map layout as per template if any and print the map.	**2**	**5**	**-**	**-**
PC17. Create map layout in software as per project map template.	2	3	-	-
PC18. Create pdf file for printing in large format printers.	-	2	-	-
QA/QC of printed map	**2**	**1**	**-**	**-**
PC19. Examine the hard copy of map for omissions and mistakes. Check cartographic details and overlapping symbols, text placement and spelling correction etc.	2	1	-	-
Create thematic map	**2**	**3**	**-**	**-**
PC20. Create a large-scale thematic map as per project requirement.	2	3	-	-
Transform Datum from one datum to another datum.	**2**	**2**	**-**	**-**
PC21. Perform datum transformation using 7 parameters.	2	2	-	-
Convert co-ordinate from one system to another.	**2**	**2**	**-**	**-**
PC22. Perform co-ordinate transformation from one system to another.	2	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	**50**	**50**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	SOI/AAS/N0107
NOS Name	Ensure GIS maps are produced as per design parameters and templates.
Sector	Aerospace and Aviation
Sub-Sector	
Occupation	Geo-Spatial Surveying Supervisor
NSQF Level	5
Credits	4
Version	1.0
Last Reviewed Date	18/07/2025
Next Review Date	18/07/2028
NSQC Clearance Date	18/07/2025

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. identify employability skills required for jobs in various industries

PC2. identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.

PC4. follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC5. recognize the significance of 21st Century Skills for employment

PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC7. use basic English for everyday conversation in different contexts, in person and over the telephone

PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English

PC9. write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

PC10. understand the difference between job and career

PC11. prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings

PC13. work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC14. communicate and behave appropriately with all genders and PwD

PC15. escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC16. select financial institutions, products and services as per requirement

PC17. carry out offline and online financial transactions, safely and securely

PC18. identify common components of salary and compute income, expenses, taxes, investments etc

PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC20. operate digital devices and carry out basic internet operations securely and safely

PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively

PC22. use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research

PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion

PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

PC26. identify different types of customers
PC27. identify and respond to customer requests and needs in a professional manner.
PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively
PC31. apply to identified job openings using offline /online methods as per requirement
PC32. answer questions politely, with clarity and confidence, during recruitment and selection
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals
KU2. various constitutional and personal values
KU3. different environmentally sustainable practices and their importance
KU4. Twenty first (21st) century skills and their importance
KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up
KU6. importance of career development and setting long- and short-term goals
KU7. about effective communication
KU8. POSH Act
KU9. Gender sensitivity and inclusivity
KU10. different types of financial institutes, products, and services
KU11. how to compute income and expenditure
KU12. importance of maintaining safety and security in offline and online financial transactions
KU13. different legal rights and laws
KU14. different types of digital devices and the procedure to operate them safely and securely
KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.
KU16. how to identify business opportunities
KU17. types and needs of customers
KU18. how to apply for a job and prepare for an interview
KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence
GS2. communicate effectively using appropriate language in formal and informal settings
GS3. behave politely and appropriately with all
GS4. how to work in a virtual mode
GS5. perform calculations efficiently
GS6. solve problems effectively
GS7. pay attention to details
GS8. manage time efficiently
GS9. maintain hygiene and sanitization to avoid infection

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Introduction to Employability Skills	**1**	**1**	**-**	**-**
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
Constitutional values – Citizenship	**1**	**1**	**-**	**-**
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
Becoming a Professional in the 21st Century	**2**	**4**	**-**	**-**
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
Basic English Skills	**2**	**3**	**-**	**-**
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Career Development & Goal Setting	**1**	**2**	**-**	**-**
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
Communication Skills	**2**	**2**	**-**	**-**
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
Diversity & Inclusion	**1**	**2**	**-**	**-**
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
Financial and Legal Literacy	**2**	**3**	**-**	**-**
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
Essential Digital Skills	**3**	**4**	**-**	**-**
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-
Entrepreneurship	**2**	**3**	**-**	**-**
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
Customer Service	**1**	**2**	**-**	**-**
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
Getting ready for apprenticeship & Jobs	**2**	**3**	**-**	**-**
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	**20**	**30**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

The trainee will be tested for his skill, knowledge and attitude during the period of course through formative assessment and at the end of the training programme through summative assessment.

a) The Continuous Assessment (Internal) during the period of training will be done by Formative Assessment Method by testing for assessment of performance listed against Performance criteria (PC). The internal assessment for each learning outcome is carried out by the concerned trainer for evaluating the knowledge and skill acquired by trainees. This internal assessment is primarily carried out by collecting evidence of competence gained by the trainees by evaluating them at work by asking questions and initiating formative discussions to assess understanding and by evaluating records and reports, and internal assessment marks are awarded to them. The methodology adopted during formative assessment will be as follows.

i. For theoretical knowledge MCQ test will be conducted to assess the knowledge of trainee against PC.

ii. For practical skill the trainee will be assessed by the trainer during practical exercise against PC

b) The training institute /NIGST will maintain individual trainee portfolio as detailed in assessment guideline. The marks of internal assessment will be as per the formative assessment template provided on www.bharatskills.gov.in

c) The final assessment will be in the form of summative assessment. The learning outcome and assessment criteria weightage will be basis for setting question papers for final assessment.

i. One Theoretical Exam (i.e. MCQ paper of 100 nos. of questions as per weightage of each NOS) with questions of medium to difficult level will be conducted. The duration will be 2 hours.

ii. One practical exam will be conducted in which participants will be provided with one of the field/Lab exercises from various NOS to perform. The trainee will be given one day (8 hrs.) to complete the field exercise and submit the results.

Sl. No. Type of assessment (Subject) Marks

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1 Formative Assessment

(i) Theory (including Employability skills) 400

(ii) Practical 350

2 Summative Assessment

(iii) Theory 100

(iv) Practical 100

TOTAL: 950

PASS REGULATION : The minimum pass percent for Formative assessment and Summative assessment are 70% . There will be no Grace marks.

Testing and certifications for the course: NIGST carries out the assessment and issues Certificates duly following the norms and guidelines issued by the NCVET.

Note: For complete assessment guideline refer the Qualification file.

Minimum Aggregate Passing % at QP Level : 70

(**Please note**: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Minimum Passing % at NOS Level: 70

(**Please note**: A Trainee must score the minimum percentage for each NOS separately as well as on the QP as a whole.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SOI/AAS/N0101.Supervise GNSS-based horizontal control surveys ensuring accuracy across all methods.	50	50	-	-	100	15
SOI/AAS/N0102.Oversee Total Station surveys for horizontal control and detail mapping.	50	50	-	-	100	10
SOI/AAS/N0103.Monitor vertical control surveys conducted using Digital Levels.	50	50	-	-	100	10
SOI/AAS/N0104.Coordinate and validate field data collection activities.	50	50	-	-	100	5
SOI/AAS/N0105.Supervise cadastral surveys ensuring spatial data accuracy.	50	50	-	-	100	10
SOI/AAS/N0106.Manage 3D data generation using UAV, photogrammetry, and LiDAR basics.	50	50	-	-	100	20
SOI/AAS/N0107.Ensure GIS maps are produced as per design parameters and templates.	50	50	-	-	100	20
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
Total	**370**	**380**	**-**	**-**	**750**	**100**

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
UAV	Unmanned Aerial Vehicle
ETS	Electronic Total Station
GNSS	Global Navigation Satellite System
ETS	Electronic Total Station
UTM	Universal Transverse Mercator
BM	Bench Mark
TBM	Temporary Bench Mark
GCP	Ground Control Point
RL	Reduced Level
DL	Digital Level
RL	Reduced Level
MSL	Mean Sea Level
EGM	Earth Gravitational Mode
BM	Bench Mark
DL	Digital Level
RL	Reduced Level
MSL	Mean Sea Level
EGM	Earth Gravitational Mode

BM	Bench Mark
GIS	Geographic Information System
UAV	Unmanned Aerial Vehicle
GIS	Geographic Information System
UAV	Unmanned Aerial Vehicle
DSM	Digital Surface Model
DEM	Digital Elevation Model
ORI	Ortho Rectified Image
ALS	Airborne Laser Scanning
LiDAR	Light Detection and Ranging
GSD	Ground Sampling Distance
GIS	Geographical Information System
ORI	Ortho Rectified Image
SDMS	Spatial Data Model Structure
RMSE	Root Mean Square Error

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
GNSS	A Global Navigation Satellite System (GNSS) is a space geodesy technique that provides autonomous geospatial positioning with global coverage.
Total Station	An electronic/optical instrument used in modern surveying that combines an electronic theodolite and electronic distance meter (EDM) to measure horizontal and vertical angles and distances.
Traverse	A series of connected survey lines whose lengths and directions are measured to establish control points.
Control Point	A fixed point on the ground with known coordinates used as a reference in surveying and mapping.
Detail Survey	A survey conducted to locate and map all physical features such as buildings, roads, boundaries, trees, poles, and other objects.
Horizontal Control	The network of points used to establish accurate horizontal positioning (X and Y coordinates).
Benchmark (BM)	A permanent reference point with a known elevation.

Digital Level	An electronic levelling instrument that automatically reads a bar-coded staff to determine height differences.
Levelling	The process of determining the relative heights or elevations of different points on the ground.
Collimation Error	An error caused when the line of sight of the levelling instrument is not perfectly horizontal.
Line of Sight	The straight line from the instrument to the staff or target being observed.
Vertical Datum	A reference surface used to measure height, such as Mean Sea Level.
Digital Level	An electronic levelling instrument that automatically reads a bar-coded staff to determine height differences.
Levelling	The process of determining the relative heights or elevations of different points on the ground.
Collimation Error	An error caused when the line of sight of the levelling instrument is not perfectly horizontal.
Line of Sight	The straight line from the instrument to the staff or target being observed.
Vertical Datum	A reference surface used to measure height, such as Mean Sea Level.
Cadastral Map	A map used for cadastral mapping showing land parcels and used for creation, updating and relaying of survey details
DSM (Digital Surface Model)	A model generated from drone/photogrammetry data representing the surface of the earth including objects
DEM (Digital Elevation Model)	A model representing the elevation of the earths surface generated from photogrammetric/UAV data
ORI (Ortho Rectified Image)	An orthorectified image generated after photogrammetric and drone data processing.
LiDAR	Technology based on laser ranging for data collection
Georeferencing	Process of assigning known coordinates to a raster or ORI
Raster Data	Image/satellite/ORI used as input dataset
Vector Data	Digitized GIS features extracted from raster
Spatial Database	Database created for storing GIS data